

High-mode dominance and the finite Schur block

PRIME.RDAGGER.HIGHMODE_DOMINANCE.01 — round 477

Stefan Hamann

August 31, 2026

Abstract

The compact-window Weil form at fixed L is a finite program: high-mode dominance (classical) plus a finite Schur block. The frequency pairing is $A = (2\pi)^{-1} \int \sigma_A(t) |\hat{h}(t)|^2 dt$ with $\sigma_A(t) = \operatorname{Re} \psi(\frac{1}{4} + it/2) - \log \pi$, identical to the r475/r476 u -space symbol (interval match at six test frequencies). Prime nodes run to e^{2L} , not e^L . A crude high-mode threshold $t_0(L)$ is sealed at $L = 0.3, 0.8, 1.0$. The crude coupling bound does not close the Schur complement; the direct 2×2 of (constant, first Dirichlet) is positive definite at all three calibration points. Yoshida/Bombieri $L = 0.3$ ($P \equiv 0$) passes. Not $\lambda_*(L) \geq 0$. Verdict ARCHITECTURE_STANDS(0.3, 0.8, 1.0). No RH claim.

Status. ARCHITECTURE_STANDS(0.3, 0.8, 1.0). No promotion of an RH claim.

Claim boundary. Nothing here is $\lambda_*(L) \geq 0$. The high-mode lemma is for an ideal frequency cutoff. The crude Schur bound $\|Q_{\ell h}\| \leq 4 \sinh L + 2S_{\text{eff}}$ is valid and useless (negative at the 2×2). Direct 2×2 positivity is not the full low-mode block. No L^* claim, no $R^\dagger$ claim, no RH claim, no anti-RH claim.

1 Frequency form

After r461/r471, $\operatorname{supp} h \subset [-L, L]$, $g = h * \tilde{h}$, $\operatorname{supp} g \subset [-2L, 2L]$, $Q_W = A - P + \Pi$ with $P(g) = \sum_{n \leq e^{2L}} 2\Lambda(n)n^{-1/2}g(\log n)$ (scramble-sensitive) and Π of rank at most two. Plancherel $\hat{h}(t) = \int h(x)e^{itx} dx$, $\|h\|_2^2 = (2\pi)^{-1} \int |\hat{h}(t)|^2 dt$:

$$A(g) = \frac{1}{2\pi} \int_{\mathbb{R}} \sigma_A(t) |\hat{h}(t)|^2 dt,$$

Lemma 1 (Symbol identity). $\sigma_A(t) = \operatorname{Re} \psi(\frac{1}{4} + it/2) - \log \pi$, equal to the r476 Bose series $-(\gamma + \log \pi) + \sum_k 2[(2k+2)^{-1} - \alpha_k/(\alpha_k^2 + t^2)]$, $\alpha_k = 2k + \frac{1}{2}$. Sealed overlap at $t = 0, 1, 2, 5, 10, 20$. In particular $\sigma_A(0) = -(\gamma + \log \pi) - (\pi/2 + 3 \log 2) \in [-5.37219, -5.37218]$ and σ_A is even and increasing on $[0, \infty)$.

2 High-mode lemma

On the ideal subspace $\{\hat{h} = 0 \text{ on } |t| < t_0\}$,

$$\frac{Q_W(h)}{\|h\|_2^2} \geq \sigma_A(t_0) - 2S_{\text{eff}}(L) - 4 \sinh L,$$

with $S_{\text{eff}} = \sum_{n \leq e^{2L}} \Lambda(n)n^{-1/2}$ exact and $|\Pi| \leq 4 \sinh L \|h\|_2^2$ from $\hat{h}(\pm i/2) = \langle h, e^{\mp x/2} \rangle$.

Lemma 2 (t_0 and c). Fix $c_\star = 0.01$. If $\sigma_A(t_0) \geq 2S_{\text{eff}} + 4 \sinh L + c_\star$ then $Q_W \geq c_\star \|h\|_2^2$ on that ideal subspace. Sealed t_0 (upper ends): $t_0(0.3) = 22$ (Bose, $\sigma_{10} = 1.2517 \geq 1.2281$), $t_0(0.8) = 4300$ (log-bound $\log |z| - \log \pi - 1/t$, calibrated at $t = 10, 20, 40$), $t_0(1.0) = 2.44 \cdot 10^5$ (same bound). At $L = 0.3$ one has $S_{\text{eff}} = 0$.

3 Schur reduction

Split frequency at t_0 : $Q_W = [Q_{\ell\ell}, Q_{\ell h}; Q_{h\ell}, Q_{hh}]$. A is diagonal in frequency; P and Π couple the blocks. If $Q_{hh} \succeq c_\star$ then $Q_W \succeq 0 \iff Q_{\ell\ell} - Q_{\ell h} Q_{hh}^{-1} Q_{h\ell} \succeq 0$. The crude operator bound $\|Q_{\ell h}\| \leq 4 \sinh L + 2S_{\text{eff}}$ makes the Schur correction larger than $Q_{\ell\ell}$ at every calibration point (YB crude Schur ≈ -125).

Lemma 3 (Direct 2×2). *On $\text{span}\{1, \cos(\pi x/(2L))\}$ the normalized Gram is positive definite: YB Schur $\in [4.673 \cdot 10^{-2}, 4.930 \cdot 10^{-2}]$, $L = 0.8$ Schur $\in [7.414 \cdot 10^{-2}, 7.507 \cdot 10^{-2}]$, $L = 1.0$ Schur $\in [3.553 \cdot 10^{-2}, 3.804 \cdot 10^{-2}]$. Constant-mode Rayleigh $r(1_{[-L,L]}) = 0.1584, 0.08027, 0.05334$ at the three L .*

4 Sealed balance

Probe `highmode_probe.py`, SPEC_SHA 57fd2de61c6fdcc9. Smoke 18/18 in 0.25 s; full 28/28 in 0.66 s. YB-gate PASS.

L	t_0	$r(1)$	$r(h_0)$	Schur 2×2	crude Schur	type
0.30	22	0.1584	$1.16 \cdot 10^{-2}$	$4.67 \cdot 10^{-2}$	-125	STANDS
0.80	4300	0.0803	$1.84 \cdot 10^{-4}$	$7.41 \cdot 10^{-2}$	$-2.0 \cdot 10^5$	STANDS
1.0	$2.44 \cdot 10^5$	0.0533	$1.10 \cdot 10^{-4}$	$3.55 \cdot 10^{-2}$	$-8.9 \cdot 10^5$	STANDS

5 Scaling forecast

$t_0 \sim 2\pi \exp(2S_{\text{eff}}(L) + 4 \sinh L)$ and $N \sim 2Lt_0/\pi$. S_{eff} grows like a truncated prime-power Mellin sum up to e^{2L} . At $L_{16} = 2.773$, $S_{\text{eff}} \approx 29.14$, $4 \sinh L \approx 31.88$, threshold ≈ 90 , hence $t_0 \sim 10^{39}$. With the crude P -bound the full block is not a practical computation. Sharpening $|g(\log n)|$ on high modes (oscillation, not $\|h\|_2^2$) turns L_{16} back into a computing question. The architecture itself does not grow with k or $L \rightarrow \infty$.

6 What this does not do

It does not prove $\lambda_*(L) \geq 0$ at any L . The ideal high-mode subspace meets compact support only at $h = 0$. The 2×2 is not the full Paley–Wiener block of width t_0 . No cofinal quantifier. No RH claim.